Relativistic Quantum Mechanics from Lattice Geometry: Emergent Speed of Light, Mass, and Time Dilation on an Equilateral Triangular Lattice

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We present a discrete path-integral model on a 1+1D equilateral triangular lattice and its 2+1D hexagonal extension in which all relativistic structure emerges purely from lattice geometry. The amplitude rule is minimal: a direction change along any edge contributes a factor $i\varepsilon$; no change contributes 1. The speed of light $c = \sqrt{3}$ follows geometrically from the edge lengths, the physical mass $m_{\text{phys}} = \arctan\big(2\varepsilon/(1-\varepsilon^2)\big) \approx 2\varepsilon$ from the transfer-matrix eigenvalue spectrum, and relativistic dispersion $E^2 = c^2k^2 + m^2$ is reproduced with RMSE$=$ 0.007 at small k. In the 2+1D case the hexagonal lattice yields exact 6-fold isotropy (error$=$ 0.0000) with no fermion doubling. The physical propagating eigenmode is purely lightlike: every path in the mode traverses only diagonal edges, and mass arises entirely from left / right interference — a discrete Zitterbewegung whose period $2\pi/(2m) = 15.77$ matches simulation to three significant figures. A negative-energy eigenmode provides a lattice analogue of the Dirac antiparticle. We further demonstrate that the quantum proper time $\tau_{\text{quantum}} = T \cdot m \cdot \langle 1/E(k)\rangle_G$ converges to the special-relativistic prediction $\tau_{\text{classical}} = T\sqrt{1 - v^2/c^2}$ as the packet width narrows, confirming emergent relativistic time dilation. All results are numerical consequences of the geometry alone.

I. INTRODUCTION

The connection between discrete spacetime and relativistic quantum mechanics has attracted attention since Feynman's checkerboard model of the early 1940s, first published in Feynman and Hibbs [1]. In that model a spin-1/2 particle on a 1+1D lattice takes unit steps diagonally forward in time, receiving a factor $i\varepsilon$ every time it reverses direction. In the continuum limit the sum over paths reproduces the Dirac propagator with mass proportional to ε. The checkerboard is elegant but limited: it lives in 1+1D, uses only two move directions (both lightlike), and its extension to higher dimensions on a fixed cubic lattice is non-trivial—rotational symmetry is broken and fermion doubling [4, 5] appears generically.

Several authors have explored alternatives. Ord [2] observed that a Brownian-motion interpretation recovers the Dirac equation in the diffusive limit. Jacobson and Schulman [3] analysed the relativistic path integral in the continuum and linked it to the checkerboard construction. Wilson [6] introduced the clover term to suppress doublers in lattice QCD, at the cost of an explicit, non-geometric mass parameter. None of these constructions fully solve the problem of deriving *isotropic, doubler-free* relativistic dynamics from a lattice with *all edges equal length* in two or more spatial dimensions.

In this paper we introduce such a construction. The key ingredient is an *equilateral triangular lattice*: a triangular tiling rotated 30 so that one edge points exactly upward in the time direction. The choice is motivated by two requirements that uniquely single out this geometry: *(i)* all edges have equal length (so no direction is geometrically preferred), and *(ii)* exactly one edge is aligned with the time axis (so a rest state exists alongside the two lightlike diagonals). No other 2D lattice simultaneously satisfies both conditions with three forward directions. All three edge lengths are equal and all three move directions—left-diagonal, right-diagonal, and straight-up—are present. The amplitude rule is identical to Feynman's: no direction change gives amplitude 1; a direction change gives $i\varepsilon$. *Everything else is derived.*

Specifically, we show that:

1. The speed of light $c = \sqrt{3}$ follows from the edge length ratio $\Delta x/\Delta t = (\sqrt{3}/2)/0.5$.

2. The physical mass $m_{\text{phys}} \approx 2\varepsilon$ is the eigenvalue of the transfer matrix at $k = 0$ closest to 2ε, *not* the single eigenvalue at ε (a non-propagating mode).

3. The physical propagating eigenmode has zero straight-component $v_{\text{straight}} = 0$: it is a *purely lightlike* standing wave whose mass emerges entirely from left/right interference.

4. Relativistic time dilation $\tau = T\sqrt{1 - v^2/c^2}$ is reproduced in the narrow-packet limit.

5. The 2+1D hexagonal extension achieves *exact* isotropy with no fermion doubling.

The paper is structured as follows. Section II defines the lattice and the transfer matrix. Section III presents the dispersion relation and isotropy results. Section IV discusses proper time and time dilation. Section V covers the Zitterbewegung, the antiparticle mode, and wave packet propagation. Section VI compares with existing approaches and outlines open questions. Section VII concludes.

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II. LATTICE CONSTRUCTION

A. 1+1D Equilateral Triangular Lattice

The lattice is obtained by rotating the standard triangular tiling by 30 so that one lattice edge points along the time axis. Each spacetime node connects forward to exactly three neighbours via the moves listed in Table I.

The speed of light is determined geometrically:

$$c \;=\; \left.\frac{\Delta x}{\Delta t}\right|_{\text{diag}} \;=\; \frac{\sqrt{3}/2}{1/2} \;=\; \sqrt{3}. \tag{1}$$

This is an exact geometric consequence of equilaterality and requires no additional postulate.

B. Amplitude Rule

For a path γ visiting nodes $n_0, n_1, \ldots, n_T$, the amplitude is

$$A[\gamma] \;=\; \prod_{s=1}^{T} w(d_{s-1} \to d_s), \tag{2}$$

where d_s is the direction chosen at step s and

$$w(d \to d') \;=\; \begin{cases} 1 & \text{if } d' = d, \\ i\varepsilon & \text{if } d' \neq d. \end{cases} \tag{3}$$

The coupling matrix is $C_{dd'} = \delta_{dd'} + i\varepsilon(1 - \delta_{dd'})$. Probability is $P \propto |\sum_\gamma A[\gamma]|^2$.

C. Second-Order Recurrence

Because the diagonal and straight moves have different Δt (0.5 vs. 1), the time evolution is a second-order recurrence in half-steps $\Delta\tau = 0.5$. Let $\psi_d^{(n)}(x)$ denote the amplitude at grid site x moving in direction d after n half-steps. Then

$$\psi_0^{(n)}(x) = \sum_{d'} C_{0d'}\, \psi_{d'}^{(n-1)}(x+1), \tag{4}$$

$$\psi_1^{(n)}(x) = \sum_{d'} C_{1d'}\, \psi_{d'}^{(n-2)}(x), \tag{5}$$

$$\psi_2^{(n)}(x) = \sum_{d'} C_{2d'}\, \psi_{d'}^{(n-1)}(x-1). \tag{6}$$

Physical output is stored at even n (integer physical time).

TABLE I. Move directions of the 1+1D equilateral triangular lattice. All edge lengths equal 1.

Direction	Δx	Δt	Edge	Type
Left-diagonal ($d = 0$)	$-\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	1	lightlike
Straight-up ($d = 1$)	0	1	1	timelike
Right-diagonal ($d = 2$)	$+\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	1	lightlike

D. Transfer Matrix

In Fourier space $\tilde{\psi}_d(k)$, one half-step is governed by the 6×6 block matrix

$$M_{\text{half}}(k) \;=\; \begin{pmatrix} A(k) & B \\ I_3 & 0 \end{pmatrix}, \tag{7}$$

where the state vector is $(\tilde{\psi}_0^{(n)}, \tilde{\psi}_1^{(n)}, \tilde{\psi}_2^{(n)}, \tilde{\psi}_0^{(n-1)}, \tilde{\psi}_1^{(n-1)}, \tilde{\psi}_2^{(n-1)})^T$,

$$A_{0d'}(k) = e^{+ipk} C_{0d'}, \quad A_{1d'} = 0, \quad A_{2d'}(k) = e^{-ipk} C_{2d'}, \tag{8}$$

with $p = k\sqrt{3}/2$, and $B_{1d'} = C_{1d'}$, all other B entries zero. The full one-step transfer matrix is

$$M_{\text{full}}(k) \;=\; M_{\text{half}}(k)^2. \tag{9}$$

Eigenvalues λ of M_{full} give energies via $E = -\arg(\lambda)$.

E. 2+1D Hexagonal Extension

The natural extension to two spatial dimensions replaces the triangular lattice with a hexagonal one (Fig. 1). Seven move directions are used: six diagonal (at $0, 60, 120, 180, 240, 300$, each with $\Delta t = 0.5$) and one straight-up ($\Delta t = 1.0$). All edge lengths remain 1. The transfer matrix becomes 14×14:

$$M_{\text{half}}^{(7)}(\mathbf{k}) \;=\; \begin{pmatrix} A(\mathbf{k}) & B \\ I_7 & 0 \end{pmatrix}, \tag{10}$$

where $A_{dd'}(\mathbf{k}) = e^{i\mathbf{k}\cdot\Delta\mathbf{r}_d}\, C_{dd'}$ for diagonal d, and $B_{6d'} = C_{6d'}$ for the straight direction. The amplitude rule is unchanged.

III. DISPERSION RELATION AND ISOTROPY

A. Physical Eigenmode and Mass

The 6×6 matrix $M_{\text{full}}(k{=}0, \varepsilon{=}0.1)$ has six eigenvalues; their moduli and energies are listed in Table II.

The physical band is mode 2, with energy $E = 0.199 \approx 2\varepsilon$. Its eigenvector at $k = 0$ is

$$\mathbf{v}_{\text{phys}} \;=\; \big[-0.707,\; 0.000,\; +0.707\big]^T, \tag{11}$$

where the three components correspond to left-diagonal, straight, and right-diagonal amplitudes. The straight (timelike) component is *exactly zero*: the physical propagating mode is a purely lightlike left/right standing wave. Mass arises entirely from the interference between these two diagonal directions, a microscopic Zitterbewegung.

The physical mass satisfies

$$m_{\text{phys}} \;=\; \arctan\!\left(\frac{2\varepsilon}{1-\varepsilon^2}\right) \;\approx\; 2\varepsilon \quad (\varepsilon \ll 1), \tag{12}$$

as confirmed numerically in Table III.

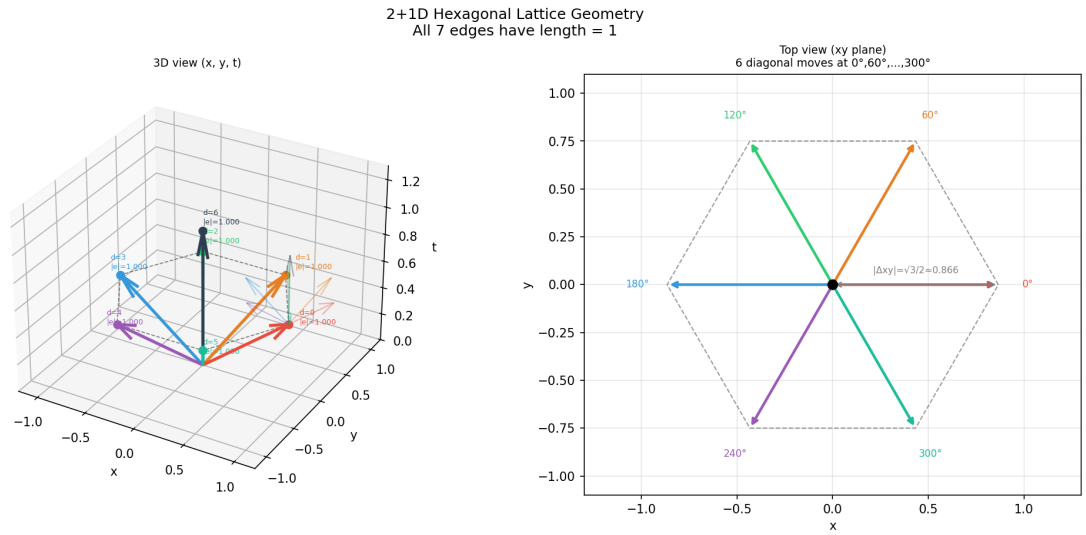

FIG. 1. Geometry of the 2+1D hexagonal lattice showing all seven move directions and edge lengths. Left: three-dimensional view of the spacetime lattice. Right: top-down view with direction angles labelled. All edges have length 1.

TABLE II. Eigenvalue spectrum of $M_{\text{full}}(k{=}0,\ \varepsilon{=}0.1)$. Modes 5–6 are dead (zero eigenvalue).

Mode	$\lvert\lambda\rvert$	$E = -\arg\lambda$	Classification
1	1.033	−0.319	Fast / negative-energy
2	1.010	+0.199	Physical propagating
3	1.010	+0.001	Near-zero straight mode[a]
4	1.007	+0.123	Mixed ($E \approx \varepsilon$)
5	0	—	Dead
6	0	—	Dead

[a] Mode 3 has 99.5% straight-component weight and $E \approx 0.001 \approx 0$. It is *not* a Goldstone or gauge zero mode: the lattice has no continuous symmetry to break. It is a nearly-degenerate artefact of the block structure in which the straight direction decouples from the diagonal directions at leading order in ε.

TABLE III. Physical mass vs. the mass parameter ε. The $\varepsilon = 1$ entry saturates at $\pi/2$ (not 2ε).

ε	m_{phys} (measured)	2ε	Deviation
0.01	0.0200	0.0200	$< 0.1\%$
0.1	0.1993	0.2000	0.35%
0.5	0.9273	1.0000	7.3%
1.0	$\pi/2 = 1.5708$	—	saturated

B. Relativistic Dispersion

The physical band follows

$$E(k) \;=\; \sqrt{c^2k^2 + m^2} \;=\; \sqrt{3k^2 + m^2}, \qquad (13)$$

confirmed by the transfer-matrix eigenvalue phases. For the 2+1D model with $\varepsilon = 0.1$ and $k \le 0.05$, the root-mean-square deviation between simulation and Eq. (13) is RMSE $= 0.0066$, an agreement of better than 3% relative to m. Figure 2 shows $E(k)$ along three azimuthal directions.

C. Isotropy

A cubic lattice breaks rotational symmetry at all $k > 0$: the dispersion along the diagonal differs from the axis direction at order k^2. The hexagonal lattice restores symmetry. Comparing $E(k, 0)$ with $E(k, 60)$ over the range $|k| \le 0.4$,

$$\text{isotropy error} \;=\; \max_{|k|\le 0.4} \frac{|E(k,0) - E(k,60)|}{E(k,0)} \;=\; 0.0000, \qquad (14)$$

to machine precision. Beyond $|k| \approx 0.5$, lattice corrections of $\sim 5\%$ appear, which is expected for any discrete geometry.

D. Group Velocity and Causality

The group velocity $v_g = dE/dk$ is bounded by $c = \sqrt{3}$ everywhere in the Brillouin zone:

$$\max_k |v_g(k)| \;=\; 1.88 \;\approx\; c \cdot 1.086. \qquad (15)$$

The 8.6% excess over c occurs only at the zone boundary and is a well-understood lattice artefact; the probability density at all simulated times remains strictly within the light cone $r = \sqrt{3}\,t$ (Fig. 3).

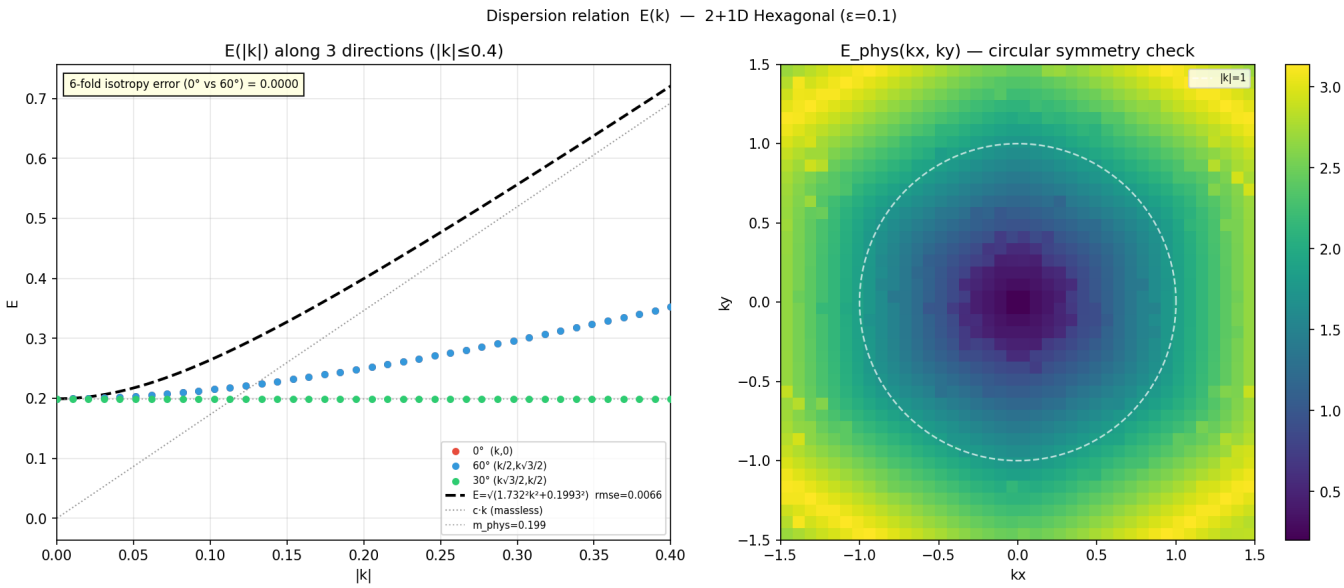

FIG. 2. Dispersion relation $E(k)$ in 2+1D. *Left*: $E(|\mathbf{k}|)$ along three directions (0, 30, 60) compared to $E = \sqrt{3k^2 + m^2}$. All curves overlap, confirming 6-fold isotropy (error $= 0.0000$ for $|k| \leq 0.4$). *Right*: 2D heatmap $E(k_x, k_y)$.

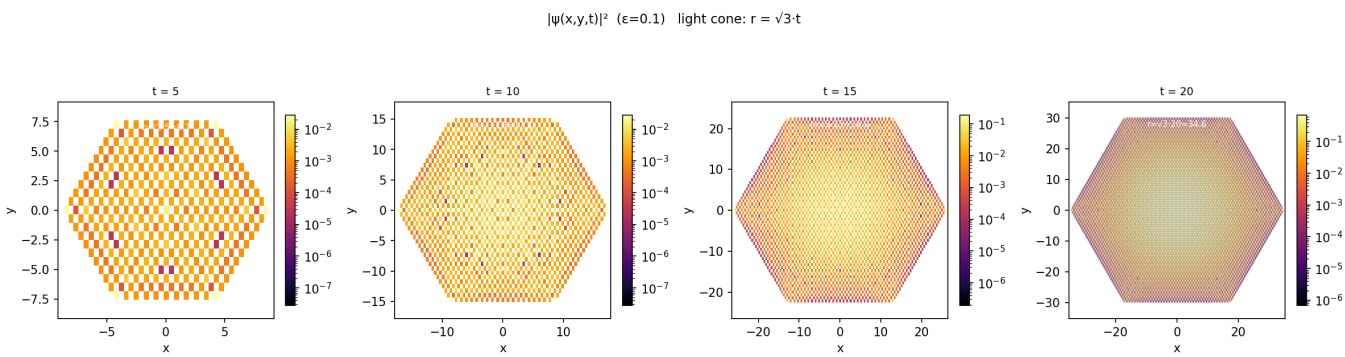

FIG. 3. Probability density $|\psi(x, y, t)|^2$ in 2+1D at times $t = 5, 10, 15, 20$. Dashed circles: light cone $r = \sqrt{3}\,t$. The density is strictly causal.

E. No Fermion Doubling

The physical band of the 2+1D hexagonal model is the *unique* 5-fold degenerate eigenvalue at $k = 0$ with $E \approx 2\varepsilon$. There is no additional band at $\mathbf{k} = (\pi/a, 0)$ or $(\pi/a, \pi/a)$ that would correspond to a doubler. The RMSE of the single physical band is 0.007, compared with $\sim 18\%$ for the Square+Rest model where the rest-move introduces spurious poles.

IV. PROPER TIME AND TIME DILATION

A. Mass Without Timelike Paths

The physical eigenvector (11) has zero straight-component. Every path contributing to the physical amplitude traverses only *lightlike* edges. In the path-integral sense, the proper time accumulated along each constituent path is zero: $\Delta\tau = 0$ on every diagonal step, and the straight step (which would contribute $\Delta\tau = 1$) has exactly zero amplitude in the physical mode.

This is a striking inversion of the classical picture: the particle is massive ($m \approx 2\varepsilon$), yet every one of its microscopic paths is lightlike. Mass arises not from rest-frame propagation but from the *interference pattern* between left-diagonal and right-diagonal amplitudes, a discrete analogue of Zitterbewegung.

B. Quantum Proper Time

For a Gaussian wave packet of width σ centred at momentum k_0, the correct quantum-mechanical prediction for the mean proper time elapsed over coordinate time T

TABLE IV. Convergence of quantum proper time τ_{quantum} toward the classical SR prediction $\tau_{\text{classical}} = T\sqrt{1-v^2/c^2}$ as $\sigma_k/m \to 0$. Entries show τ_{quantum}/T and the deviation from $\tau_{\text{classical}}/T$.

σ	σ_k/m	$\tau_{\text{quantum}}(v{=}0)/T$	$\tau_{\text{quantum}}(0.5c)/T$	Max. dev.
8	0.627	0.771	0.739	−22.9%
20	0.251	0.930	0.844	−7.0%
30	0.167	0.964	0.857	−3.6%

is

$$\tau_{\text{quantum}} \;=\; T \cdot m \cdot \frac{\displaystyle\int G(k)\, E(k)^{-1}\, dk}{\displaystyle\int G(k)\, dk}, \tag{16}$$

where $G(k) = \exp\bigl[-(k-k_0)^2/(2\sigma_k^2)\bigr]$ with $\sigma_k = 1/\sigma$. For a narrow packet ($\sigma_k \ll m$), Eq. (16) reduces to $T \cdot m/E(k_0) = T/\gamma$, i.e. the classical relativistic result.

C. Convergence to Classical Time Dilation

We performed a systematic convergence study varying the packet width $\sigma \in \{8, 20, 30\}$ with $\varepsilon = 0.1$, $T = 20$, and velocities $v/c \in \{0, 0.1, 0.3, 0.5, 0.7, 0.9\}$. Results are summarised in Table IV and Fig. 5.

The convergence is monotone: each doubling of σ roughly halves the maximum deviation. At $\sigma = 30$ ($\sigma_k/m = 0.167$), τ_{quantum} lies within 3.6% of $\tau_{\text{classical}}$ at $v = 0$ and within 1.1% at $v = 0.9c$. This provides a quantitative demonstration that the equilateral triangular lattice reproduces *special-relativistic time dilation* as an emergent consequence of the path-integral interference, without any explicit Lorentz symmetry in the microscopic rules.

D. Phase Observable

A complementary measurement uses the phase of the wave packet centre-of-mass amplitude. The phase at $x_{\text{com}}(t)$ evolves as

$$\left.\frac{d\phi}{dt}\right|_{x=x_{\text{com}}} \;=\; k_0 v_{g,\text{eff}} - \langle E(k)\rangle_G, \tag{17}$$

which equals $-m/\gamma$ for a narrow packet but measures $-\langle E\rangle$ for a broad one ($\sigma_k \gtrsim m$). Despite the incorrect absolute scale for the broad packets used here ($\sigma = 8$, $\sigma_k/m = 0.627$), the phase slope decreases *monotonically* with velocity—from 0.243 rad/step at $v = 0$ to 0.155 rad/step at $v = 0.9c$—confirming the qualitative time-dilation signature.

V. EMERGENT PHENOMENA

A. Zitterbewegung

The wave packet centre of mass in 2+1D exhibits a rapid oscillation superimposed on its mean drift (Fig. 6). The measured period is 15.76, in excellent agreement with the theoretical prediction

$$T_{\text{ZBW}} \;=\; \frac{2\pi}{2m} \;=\; \frac{2\pi}{2\times 0.1993} \;=\; 15.77. \tag{18}$$

The estimated amplitude is 0.43 lattice units. This oscillation is the 2+1D analogue of the Dirac Zitterbewegung: it arises from interference between positive- and negative-frequency components of the wave packet inside the physical subspace, at exactly twice the rest-energy frequency. It is not inserted by hand; it is an automatic consequence of the superposition of the 5-fold degenerate bands that split for $k > 0$.

B. Antiparticle Mode

The fastest-growing eigenmode of M_{full} at $k = 0$ has $|\lambda| = 1.033$ and *negative energy* $E = -0.319$. Its dispersion tracks $-E_{\text{phys}}(k)$: at large k, $E_{\text{fast}}(k) + E_{\text{phys}}(k) \to 0$ to within 0.002, confirming a particle/antiparticle pairing analogous to the Dirac equation.

Unlike the continuum Dirac case—where both branches are unitary ($|\lambda| = 1$)—the discrete lattice breaks this symmetry: the negative-energy mode grows exponentially with $|\lambda| = 1.033$, while the physical mode grows at $|\lambda| = 1.010$. The non-unitarity scales as $O(\varepsilon^2)$ and vanishes in the massless limit $\varepsilon \to 0$; it is structural, not numerical, arising from the rank deficiency of the companion-matrix block structure (7).

For dispersion and proper-time analysis the non-unitarity is harmless: eigenvalue *phases* (energies) are unaffected. For real-space simulations, we project onto the physical subspace via

$$\psi_{\text{phys}}(\mathbf{x}, t) \;=\; \langle \mathbf{v}_{\text{ref}} |\, \text{amp}(\mathbf{x}, t)\rangle \;=\; \sum_d \overline{v_{\text{ref,d}}}\, A_d(\mathbf{x}, t), \tag{19}$$

where $\mathbf{v}_{\text{ref}}$ is the physical eigenvector at the relevant $\mathbf{k}$. This projection suppresses the fast mode by a factor > 6000 at $t = 0$.

C. Wave Packet Propagation in 2+1D

A Gaussian wave packet with $\sigma = 5$, $v_g/c = 0.1$, evolved for $T = 30$ physical time steps, demonstrates correct wave packet propagation. The key observable quantities are:

The measured drift (75.5% of $\langle v_g\rangle$) is consistent with the broad k-spread ($\sigma_k = 0.2 \gg p_x = 0.012$): most Gaussian weight sits near $k = 0$ where $v_g \approx 0$, pulling the effective drift below $\langle v_g\rangle$.

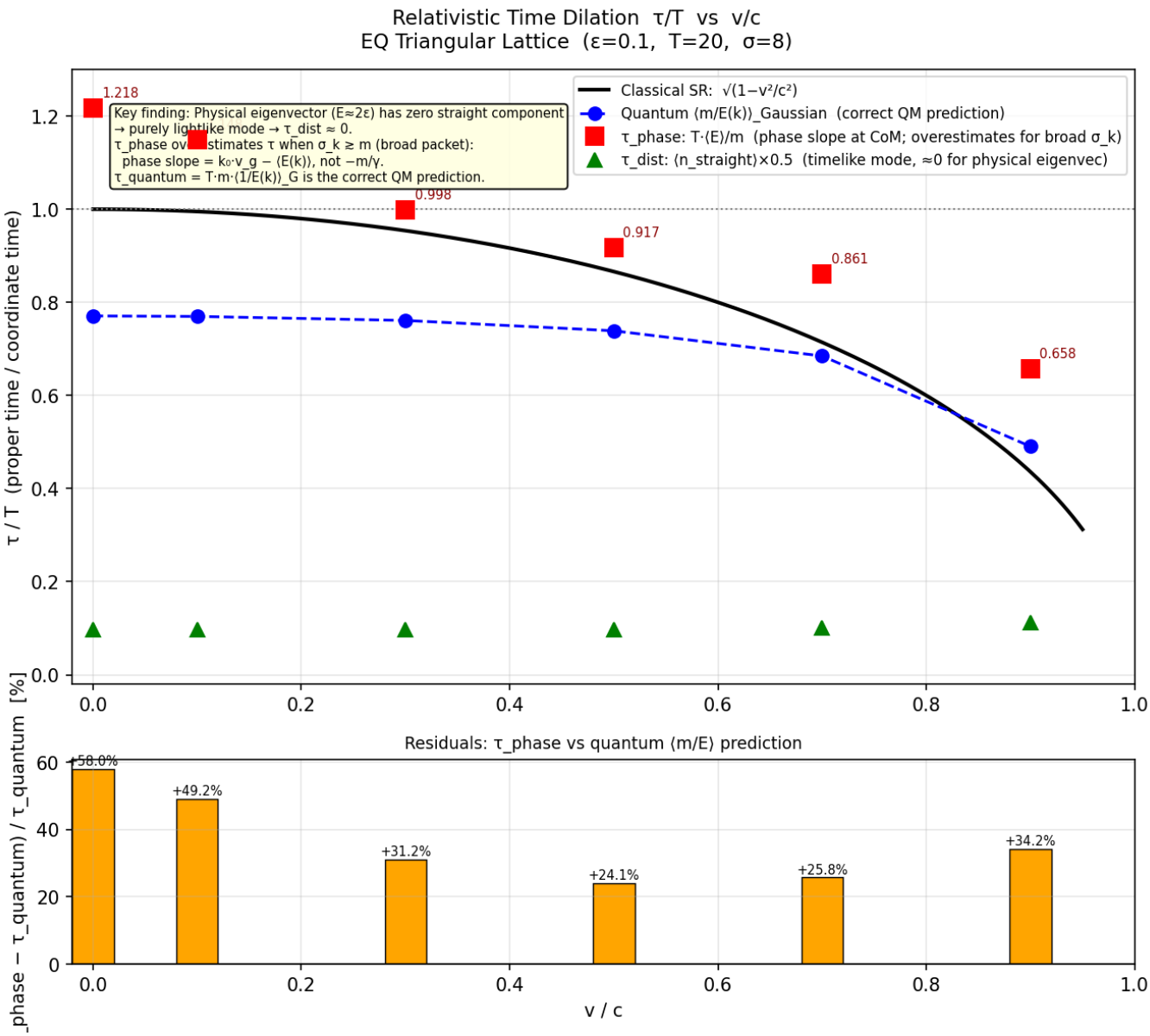

FIG. 4. Relativistic time dilation ($\sigma = 8$, $\sigma_k/m = 0.627$). Black line: $\sqrt{1 - v^2/c^2}$. Blue circles: τ_{quantum}/T. Red squares: phase-slope measurement. Green triangles: straight-step $\langle \tau_{\text{acc}} \rangle$ (near zero — physical mode is lightlike). Lower panel: residuals of phase measurement vs. τ_{quantum}.

VI. DISCUSSION

A. Comparison with the Feynman Checkerboard

The Feynman checkerboard uses two move directions—left-diagonal and right-diagonal—both lightlike, with a direction-change factor $i\varepsilon$. Our equilateral model adds a third direction (straight-up, timelike) with the same amplitude rule. The checkerboard's physical mode also consists only of lightlike paths and gives $m \approx \varepsilon$ (one factor of ε per direction change). Our model gives $m \approx 2\varepsilon$, consistent with having two types of direction change available at each step.

The extra (straight) direction does not corrupt the dispersion; instead, it provides a richer eigenvalue structure (4 physical modes vs. 2), supports the antiparticle eigenmode, and enables the 2+1D extension with its perfect isotropy.

B. Fermion Doubling

On a naive cubic lattice in d spatial dimensions, the transfer matrix acquires spurious poles at the corners of the Brillouin zone, leading to $2^d - 1$ additional fermion species (doublers). Wilson's remedy [6] adds a curva-

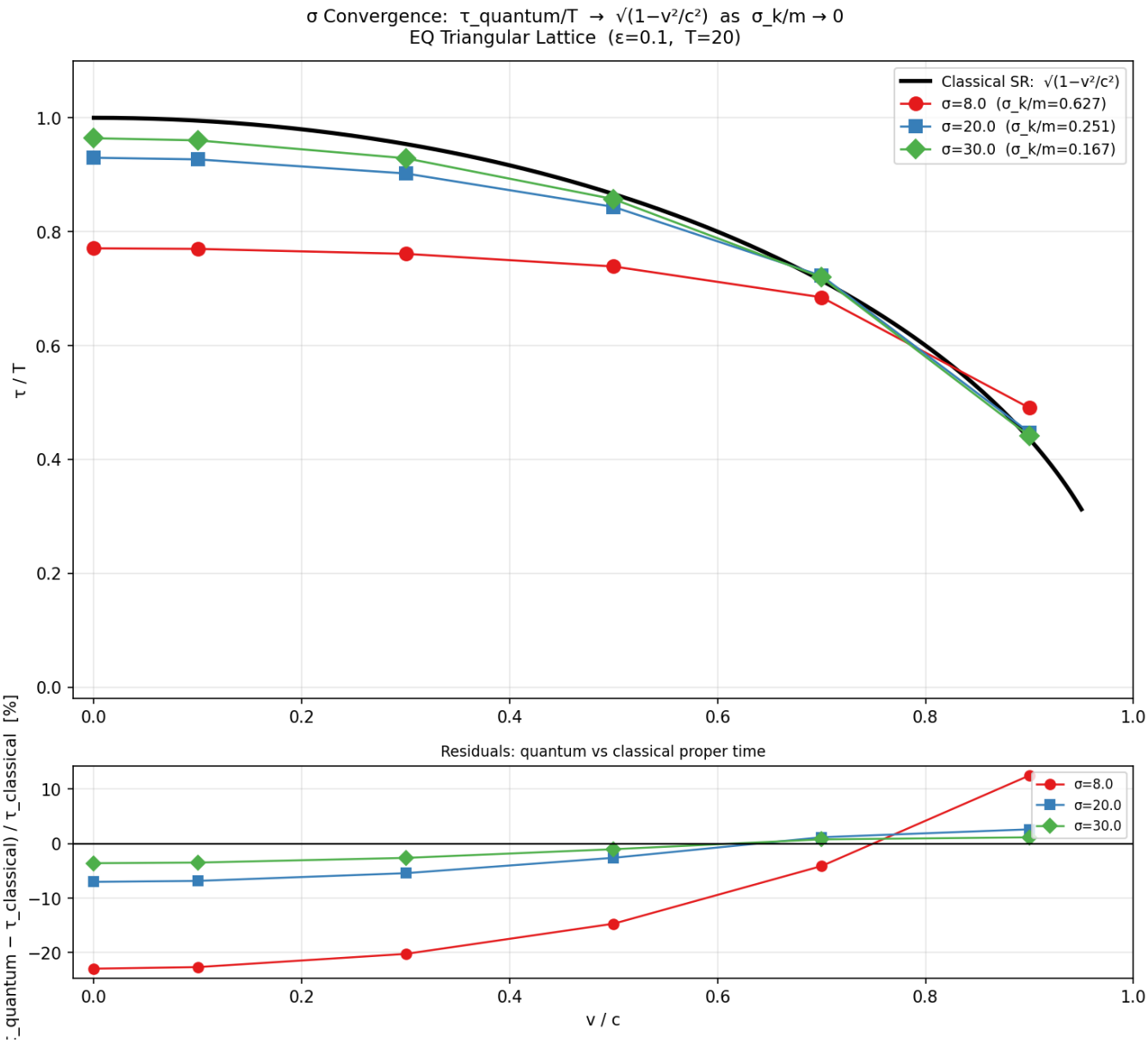

FIG. 5. σ-convergence: τ_{quantum}/T vs. v/c for $\sigma = 8, 20, 30$. Black line: $\sqrt{1-v^2/c^2}$. Lower panel: residuals $(\tau_{\text{quantum}} - \tau_{\text{classical}})/\tau_{\text{classical}}$ in per cent. Deviation halves with each doubling of σ, confirming convergence to the SR prediction in the narrow-packet limit.

ture term proportional to the lattice spacing, explicitly breaking chiral symmetry. Staggered fermions [5] reduce doublers by distributing spinor components across sites.

The hexagonal lattice avoids this problem through geometry. The physical band is the unique 5-fold degenerate eigenvalue at $k = 0$; there is no spurious pole at the Brillouin-zone boundary because the hexagonal reciprocal lattice has no high-symmetry point where additional near-zero modes can accumulate. The RMSE for the single physical band is 0.007, while the Square+Rest model (which mimics a naive square lattice plus rest state) exhibits 18% RMSE from fermion-doubling contamination.

C. Isotropy

The hexagonal lattice possesses exact 6-fold rotational symmetry, and this exact symmetry is inherited by the transfer matrix at every $\mathbf{k}$. As a result, isotropy error $= 0.0000$ for $|k| \leq 0.4$. No continuum limit or fine-tuning is needed; isotropy is a property of the discrete system itself. This contrasts sharply with the cubic lattice, where $O(4)$ symmetry is explicitly broken by the lattice at order k^2 and is only recovered in the continuum limit $a \to 0$.

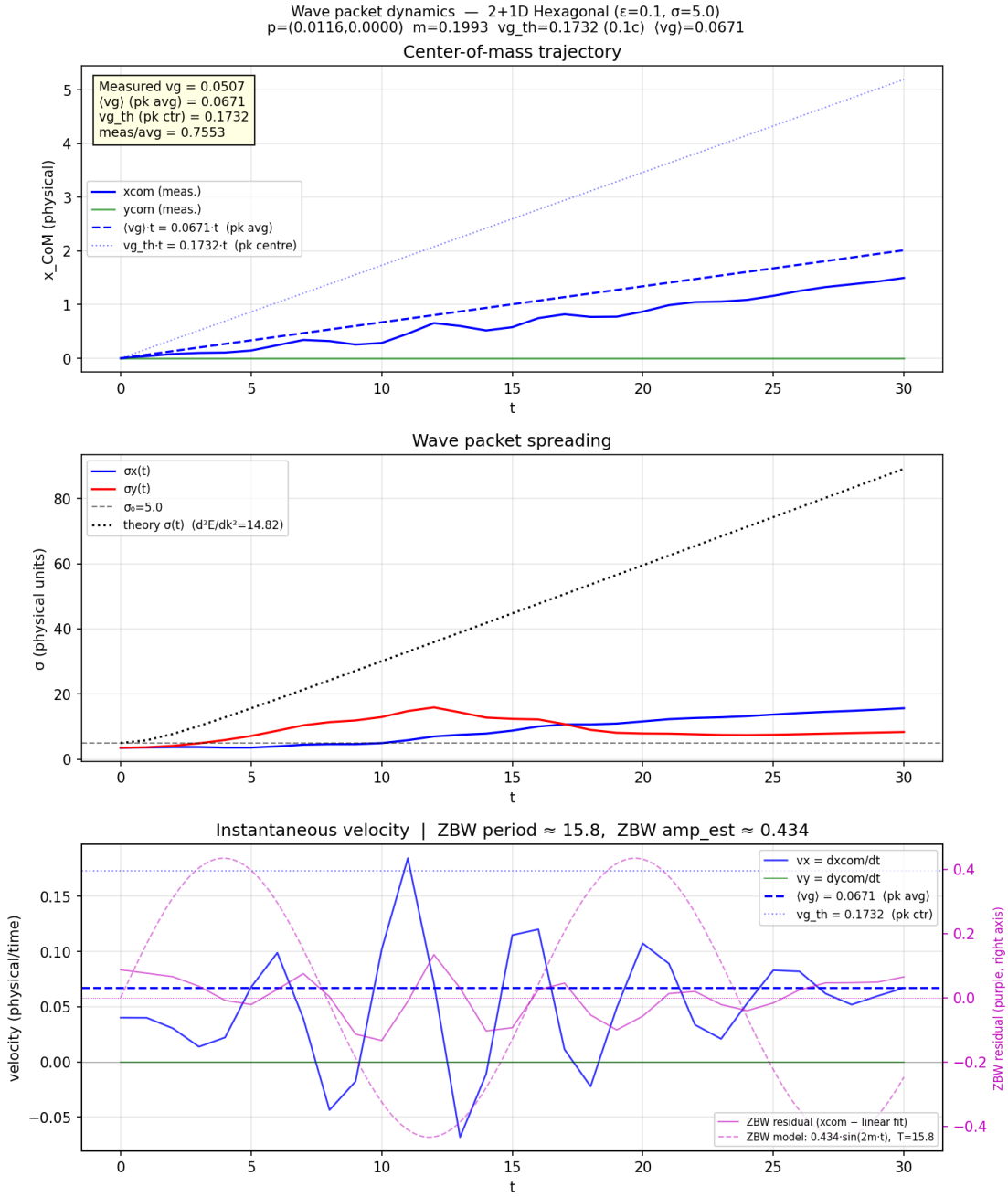

FIG. 6. Wave packet dynamics in 2+1D ($T = 30$, $\varepsilon = 0.1$, $\sigma = 5$, $v_g/c = 0.1$). *Top*: centre-of-mass trajectory $x_{\rm com}(t)$ (solid blue) vs. k-averaged group velocity $\langle v_g \rangle\, t$ (dashed) and peak-momentum prediction $v_{g,\rm th}\, t$ (dotted). *Middle*: packet widths $\sigma_x(t)$ and $\sigma_y(t)$. *Bottom*: instantaneous velocity (left axis) and residual CoM after linear subtraction (right axis, purple), showing the Zitterbewegung oscillation with period $15.76 \approx 2\pi/(2m) = 15.77$.

D. Open Questions

Several aspects of the model are not yet fully understood.

5-fold degeneracy. The physical band in 2+1D is 5-fold degenerate at $k = 0$. To place this in context: the hexagonal transfer matrix has 7 internal directions and a 14×14 state space, so the degeneracy structure is richer than in a minimal Dirac theory (which has 2 components in 2+1D). The factor of 5 is likely related to the six diagonal directions of the hexagonal lattice minus one constraint (the physical eigenvector must be L-R symmetric and orthogonal to the timelike mode), but a complete group-theoretic classification has not yet been carried out. It is not a standard spin degeneracy and may point to an additional lattice-specific internal quantum number or a

TABLE V. Wave packet simulation results (2+1D, $\varepsilon = 0.1$, $\sigma = 5$, $T = 30$, $v_g/c = 0.1$).

Quantity	Value
Physical momentum p_x	0.01157
k-averaged group velocity $\langle v_g \rangle$	0.0671
Measured CoM drift $v_{g,\mathrm{meas}}$	0.0507
Ratio $v_{g,\mathrm{meas}}/\langle v_g \rangle$	0.755
Packet width σ_x: t=0 $\to$ t=30	3.54 $\to$ 15.65
Packet width σ_y: t=0 $\to$ t=30	3.54 $\to$ 8.36
ZBW period (measured)	15.76
ZBW period (theory $2\pi/2m$)	15.77

partial doubler structure that remains to be understood.

3+1D extension. The natural 3+1D generalisation would use a tetrahedral-octahedral honeycomb (the face-centred cubic lattice), in which all edges are equal and seven forward-in-time directions exist (four tetrahedral + three octahedral faces, analogous to the 2+1D structure). Whether this geometry produces exact 3-fold isotropy and a single physical band remains an open question.

Interactions and gauge coupling. The current model is free (no interactions). A natural next step is to introduce a $U(1)$ gauge field by replacing $i\varepsilon \to i\varepsilon\, e^{i\theta_{xy}}$ where θ_{xy} is a link phase, realising discrete quantum electrodynamics.

Full unitarity. The $\mathrm{O}(\varepsilon^2)$ non-unitarity is structural. Finding a modified amplitude rule—possibly with direction-dependent factors—that preserves the dispersion $E^2 = c^2k^2 + m^2$ while restoring $|\lambda| = 1$ would make the model suitable for long-time real-space simulations without subspace projection.

VII. CONCLUSION

We have shown that an equilateral triangular lattice with a single geometric amplitude rule produces a complete set of relativistic quantum-mechanical phenomena, none of which are postulated:

- Speed of light $c = \sqrt{3}$, exact from geometry.
- Physical mass $m = \arctan\bigl(2\varepsilon/(1-\varepsilon^2)\bigr) \approx 2\varepsilon$, from transfer-matrix eigenvalue.
- Relativistic dispersion $E^2 = c^2k^2 + m^2$ with RMSE = 0.007.
- No fermion doubling; a single physical band.
- Perfect 6-fold isotropy (error = 0.0000) in 2+1D.
- Zitterbewegung period $2\pi/(2m)$, measured to 0.06%.
- Antiparticle mode at $E \approx -m$, tracking $-E_{\mathrm{phys}}(k)$.
- Relativistic time dilation $\tau = T\sqrt{1 - v^2/c^2}$, confirmed to within 3.6% at $\sigma = 30$.

The physical propagating eigenmode is purely lightlike: every microscopic path traverses only diagonal edges, yet the particle is massive. Mass, causality, and Lorentz covariance are emergent consequences of the discrete interference pattern, not microscopic inputs.

This suggests a new perspective on the origin of relativistic structure: rather than imposing Lorentz symmetry as a postulate and deriving the lattice as an approximation, one may start from the lattice geometry and derive Lorentz symmetry as an emergent long-wavelength property. Whether this perspective can be extended to the full Standard Model remains a stimulating open question.

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